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About this Course

Key Components of Llamafile

Developing Portable CLI with Cosmopolitan

✓ **Reading:** Key Terms
1 min

✓ **Video:** Building a phrase generator with cosmopolitan
3 min

📄 **Lab:** Portable CLI
10 min

✓ **Reading:** Bash Phrase Generator
5 min

✓ **Graded Assignment:** Quiz-Portable CLI with Cosmopolitan
Submitted

📄 **Reading:** Lesson Reflection
5 min

Running Llamafile

Conclusion and Next Steps

Bash Phrase Generator

Lab Goal Apply concepts from lesson by modifying, running, and expanding a Bash script to repeat customizable phrases.

Lab Description & Steps

1. Analyze the phrase generator Bash script components
2. Handle invalid counts by defaulting to 1 phrase
3. Append a counter to each phrase printed
4. Support repeating multi-line phrases
5. Extend script functionality with your own options

Reflection Questions

1. How does a case statement enable flexible CLI argument parsing?
2. When might default values be important in argument handling?
3. What is the difference between echo and printf for display?
4. Why wrap key logic in easy to reuse functions?
5. Where else could scripting facilitate portability?

Challenge Exercises

1. Add support for spacing repeated phrases apart
2. Generate phrases from a file instead of arguments
3. Create a user input mode rather than reading CLI args
4. Implement counting phrases using different methods
5. Develop and share your own creative phrase repeater scripts

Bash example

```
1  #!/bin/bash
2
3  # Function to generate phrases
4  generate_phrases() {
5      local count=$1
6      local phrase=$2
7
8      for ((i=0; i<count; i++)); do
9          echo $phrase
10     done
11 }
12
13 # Default values
14 count=1
15 phrase="Hello world"
16
17 # Read arguments
18 while [[ $# -gt 0 ]]; do
19     case $1 in
20         -c|--count)
21             count="$2"
22             shift ;;
23         -p|--phrase)
24             phrase="$2"
25             shift ;;
26         esac
27     shift
28 done
29
30 # Generate phrases
31 generate_phrases "$count" "$phrase"
```

Running it

```
1  $ ./phrase-gen.sh
2  Hello world
3
4  $ ./phrase-gen.sh -c 5
5  Hello world
6  Hello world
7  Hello world
8  Hello world
9  Hello world
10
11 $ ./phrase-gen.sh -p "Greetings" -c 3
12 Greetings
13 Greetings
14 Greetings
```

Go to next item

✓ Completed

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